



Launch a Kubernetes Cluster



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Nodes (3) [Info](#)

< 1 >

Node name	Instance type	Compute	Managed by	Created	Status
ip-192-168-11-57.ec2.internal	t3.micro	Node group	nextwork-nodegroup	2 hours ago	Ready
ip-192-168-38-45.ec2.internal	t3.micro	Node group	nextwork-nodegroup	2 hours ago	Ready
ip-192-168-50-117.ec2.internal	t3.micro	Node group	nextwork-nodegroup	2 hours ago	Ready

Node groups (1) [Info](#)

EditDeleteAdd

< 1 >

Group name	Desired size	AMI release version	Launch template	Status
nextwork-nodegroup	3	1.53.5-20251217	eksctl-nextwork-eks-cluster-nodegroup-nextwork-nodegroup (1)	Active



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Introducing Today's Project!

In this project, I will be launching and connecting to an EC2 instance, creating a Kubernetes cluster, then monitor its deployment via CloudFormation, and accessing it through an IAM access entry, and then I will be evaluating the cluster's resilience.



What is Kubernetes?

Kubernetes is a container orchestration platform which automates the deployment, scaling, and management of containers across a cluster of machines. Companies and developers use Kubernetes to run containerized applications reliably and at scale

I used eksctl to create EKS cluster. The create cluster command I ran defined the EKS cluster name, node group name, minimum and maximum number of node based on demand.

I initially ran into two errors while using eksctl. The first one was because of the absence of permission in the roles of the instance to create an EKS cluster, The second one was because of the eksctl not installed in the EC2 terminal.

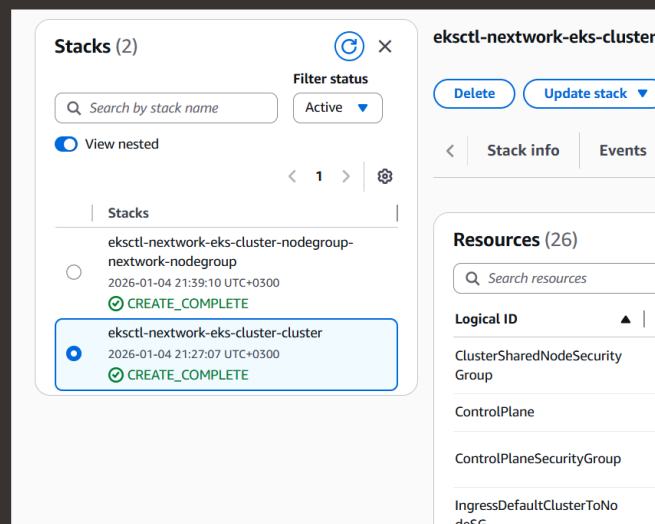
```
ec2-user@ip-172-31-14-197 ~]$ eksctl create cluster \
-name nextwork-eks-cluster \
-nodegroup-name nextwork-nodegroup \
-node-type t3.micro \
-nodes 3 \
-nodes-min 1 \
-nodes-max 3 \
-version 1.33 \
-region us-east-1
bash: eksctl: command not found
ec2-user@ip-172-31-14-197 ~]$
```



eksctl and CloudFormation

CloudFormation helped created my EKS cluster because it automated the provisioning and configuration of all required AWS resources in a consistent way. It also created VPC resources because EKS requires a properly configured networking environment like subnets, route tables, and security groups in order to enable secure communications.

After checking CloudFormation for the status of the EKS creation, there was also a second CloudFormation stack for the node group, this stack is for the instances that will run the containers. The difference between a cluster and node group is that the cluster is the entire environment that the Kubernetes manages for the containerized apps and node group is the collection of nodes.





The EKS console

I had to create an IAM access entry in order to gain permission to view my created nodes in the node group. An access entry is like a handshake that connects AWS and Kubernetes. IAM is AWS's login and permission system, while RBAC is Kubernetes's. This connects the IAM role to an RBAC role. I set it up by opening the Access tab, Under IAM access entries, I select ed Create access entry and continued till the end of the creation page.

It took 10 to 15 minutes to create my cluster.

The screenshot displays the AWS EKS console interface. The top section, titled 'Nodes (3)', shows a table of three nodes, all in a 'Ready' status. The bottom section, titled 'Node groups (1)', shows a single node group named 'nextwork-nodegroup' with a status of 'Active'.

Nodes (3)						
Filter Nodes by property or value						
Node name	Instance type	Compute	Managed by	Created	Status	
ip-192-168-11-57.ec2.internal	t3.micro	Node group	nextwork-nodegroup	2 hours ago	Ready	
ip-192-168-38-45.ec2.internal	t3.micro	Node group	nextwork-nodegroup	2 hours ago	Ready	
ip-192-168-50-117.ec2.internal	t3.micro	Node group	nextwork-nodegroup	2 hours ago	Ready	

Node groups (1)					
Node groups implement basic compute scaling through EC2 Auto Scaling groups.					
Filter node groups by property or value					
Group name	Desired size	AMI release version	Launch template	Status	
nextwork-nodegroup	3	1.33.5-20251217	eksctl-nextwork-eks-cluster-nodegroup-nextwork-nodegroup (1)	Active	



EXTRA: Deleting nodes

Did you know you can find your EKS cluster's nodes in Amazon EC2? This is because each EKS node is actually an EC2 instance running in your AWS account.

Desired size means the number of nodes you want running in your node group. Minimum and maximum sizes are helpful for scaling the nodes up and down.

When I deleted my EC2 instances the instances got replaced, This is because they were part of an EKS node group, which automatically maintains the desired number of nodes by launching new instances whenever existing ones are terminated.

☐	network-eks-...	i-05c24a270e0bff66d	✔ Running 🔍 🔍	t3.micro	✔ 3/3 checks passed View alarms +
☐	network-eks-...	i-04139d2e402fed14e	✔ Running 🔍 🔍	t3.micro	✔ 3/3 checks passed View alarms +
☐	network-eks-...	i-0e03e1f8ac8d48bcc	✔ Running 🔍 🔍	t3.micro	✔ 3/3 checks passed View alarms +



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